PAVEMENT-PINN — Master Development Tracker

Updated 2026-10-15 · branch `feature/m1-report-ownership` (HEAD `4a9b4fc` + uncommitted work) · Sources: v3 Dokumen Induk (governs), `AGENTS.md`, `pavement_pinn_v3_code.zip`. Full baseline (spec-vs-implementation matrix, 51 findings, 52 roadmap tasks, math checks): `docs/audit/v3-baseline-2026-10-15/`. This file tracks status changes since that baseline.

1. Actual state

Area	State	Evidence
Workflow, server RBAC, separation of duties, audit trail	Working, tested	229/229 `npm test`
Structural engine	RULE_BASED heuristic; strains 15×/47× below MLET on the v3 reference case	baseline §M, F-03
PI-surrogate, Monte Carlo, YOLO, SLM, benchmarks	Not built	no `inference/`, `models/`, `measurements/`
Python runtime on this machine	Blocked by machine policy	F-38 / HD-01
Cloud: RunPod	1 GPU pod (RTX PRO 4000), **EXITED**; 0 endpoints, 0 volumes; nothing of PAVEMENT-PINN deployed	`evidence/cloud-status-2026-10-15.md`
Cloud: Vercel	**Not verified**: connector not authorized in this session	—
Persistence	Local JSON file store; Supabase migration not applied	F-24

2. Phase 0 "Truthful status" — done this session

Task	Findings	Status	Closure evidence
R0-02 fake telemetry/loss/attribution	F-02, F-14, F-20, F-21	**CLOSED**	`IntelTelemetry` fields `number | null` + `measurementStatus: NOT_MEASURED`; 0 `Math.random` in src/ and server/ (guard test)
R0-03 WIM modal, scraper, simulated benchmark, IR/config export	F-04, F-13, F-31	**CLOSED**	files deleted; `/api/openvino/export-*` → 404 (api.test)
R0-09 WIM stations with plates, "FWD Validated"	F-16	**CLOSED**	`ALL_BINA_MARGA_WIM_STATIONS = []`; no segment has WIM data; badge removed
R0-04 CV confidence	F-05	**CLOSED**	confidence `null` + `modelStatus: NOT_CLASSIFIED`; fake detection box removed; history shows "belum diklasifikasi"
R0-05 verdict/order text	F-06 (text)	**CLOSED**	engine notes, alerts, narratives and offline fallbacks reworded; no "aman"/"batas ijin"/"wajib penindakan"; fallbacks no longer invent default numbers
R0-06 browser NPU claim	F-12	**CLOSED**	`deviceDetector` reports "tidak diketahui", never NPU
R0-10 PDF disclaimer	F-19	**CLOSED**	`PDF_DISCLAIMER` on every page; agency role names and money lines removed
R0-01 "PINN solver" wording	F-01	**PARTIAL**	README, engine, PDF, main screens fixed; legacy screens below still carry agency/PINN copy
R0-08 decision log	F-40	**CLOSED**	`docs/DECISION_LOG.md`: V3-D-01…18, hierarchy, D-035
R0-12 README	F-30	**CLOSED**	README "Status jujur" section
R0-07 Narogong corridor	F-15, D-03	**BLOCKED**	needs real geometry + managing authority (HD-07); not fabricated
R0-11 status label on every page	F-23	OPEN	—

New gate: `tests/unit/no-fake-claims.test.ts` (no Math.random, prohibited phrases, deleted modules, NPU claim, PDF disclaimer, determinism).

Tests changed because they encoded a P0 false claim (recorded in D-035): golden regression now pins every numeric leaf outside `telemetry` (still tolerance 0; physics numbers unchanged); telemetry jitter test → NOT_MEASURED test; export tests → 404; `/api/roads` size/hash updated (18133 → 8775 bytes, WIM data removed); narrative fallback text; two threshold tests now assert the new wording (thresholds 180/250 still pinned).

Residual Phase-0 text (P1, open as R0-01b)

Agency or "PINN" wording still in legacy screens: `PavementPinnDashboard.tsx`, `FieldEngineerTools.tsx`, `OperationsHomeView.tsx`, `OperationsPosWimMapView.tsx`, `LeafletMap.tsx` ("WIM (PUPR)"), `PinnPage.tsx`, `FwdPage.tsx`, `ExecutivePolicyView.tsx` (memo titles, "Sumber: Kementerian PUPR"). Engine fields `preventiveCostSavingsRupiahBillions`, `potholeHorizonDays`, PCI projection and `wimActionRequired` still exist in data (removal is R1-06, with the engine replacement).

3. Test results (actual, 2026-10-15)

Check	Result
`npm run lint` (`tsc --noEmit`)	exit 0
`npm test`	229 tests, 229 pass, 0 fail
`vite build`	success
`pytest`	not run: no Python runtime (F-38)

4. Remaining plan (order unchanged from baseline Appendix F)

1. R0-01b residual copy, R0-11 status labels: no blockers.
2. Phase 1–5 (`inference/` FastAPI, PyMastic 303/793 test, surrogate, Trust Gate, YOLO, SLM, benchmarks): **blocked by HD-01**. The exited RunPod pod could host *training* only; v3 §5.2 benchmarks must run on the Intel interview laptop.
3. Phase 6 role features, Phase 7 Playwright + deployment: after Phase 1.

5. Decisions needed from you

HD-01 Python machine/policy · HD-02 workflow state names · HD-03 YOLO licence (AGPL) · HD-04 transfer-function criterion · HD-05 surrogate domain (ln(d/a)) · HD-06 Supabase vs VM · HD-07 Narogong data · HD-08 commit trailer (`Assisted-by` vs Pavement-PINN co-author) · HD-09 reference-ZIP adoption / who rewrites the science core. F-52 (new, P2): RunPod pod stores its Jupyter password in plain env config; rotate it before restarting the pod.

6. Readiness checklist (AGENTS §8 Definition of Done)

- [x] No on-screen number from a constant or `Math.random` in telemetry/CV/attribution paths (CI guard green)
- [ ] Photo → gate → blur → versioned detector → scored queue
- [ ] PyMastic εt/εv + life index P10/P50/P90 with criterion
- [ ] Offline assessment + NPU brief
- [ ] Evidence page with real benchmarks
- [ ] npm + pytest + Playwright green (npm only so far)